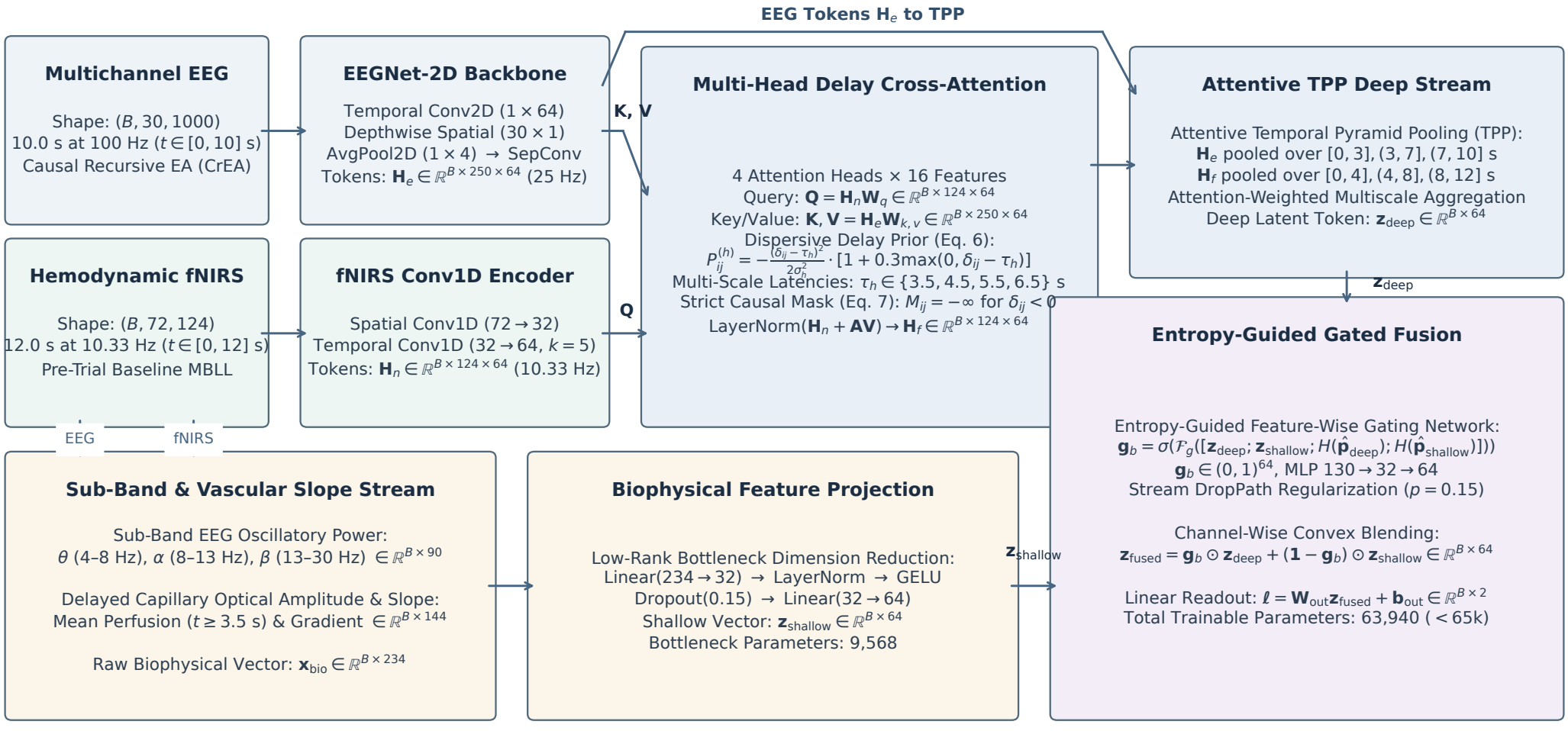

EEG Tokens $\mathbf{H}_e$ to TPP
Multichannel EEG
Shape: $(B, 30, 1000)$
10.0 s at 100 Hz ($t \in [0, 10]$ s)
Causal Recursive EA (CrEA)
EEGNet-2D Backbone
Temporal Conv2D (1×64)
Depthwise Spatial (30×1)
AvgPool2D (1×4) → SepConv
Tokens: $\mathbf{H}_e \in \mathbb{R}^{B \times 250 \times 64}$ (25 Hz)
K, V
Multi-Head Delay Cross-Attention
4 Attention Heads × 16 Features
Query: $\mathbf{Q} = \mathbf{H}_n \mathbf{W}_q \in \mathbb{R}^{B \times 124 \times 64}$
Key/Value: $\mathbf{K}, \mathbf{V} = \mathbf{H}_e \mathbf{W}_{k,v} \in \mathbb{R}^{B \times 250 \times 64}$
Dispersive Delay Prior (Eq. 6):
$P_{ij}^{(h)} = -\frac{(\delta_{ij} - \tau_h)^2}{2\sigma_h^2} \cdot [1 + 0.3\max(0, \delta_{ij} - \tau_h)]$
Multi-Scale Latencies: $\tau_h \in \{3.5, 4.5, 5.5, 6.5\}$ s
Strict Causal Mask (Eq. 7): $M_{ij} = -\infty$ for $\delta_{ij} < 0$
LayerNorm$(\mathbf{H}_n + \mathbf{AV}) \to \mathbf{H}_f \in \mathbb{R}^{B \times 124 \times 64}$
Attentive TPP Deep Stream
Attentive Temporal Pyramid Pooling (TPP):
$\mathbf{H}_e$ pooled over $[0, 3], (3, 7], (7, 10]$ s
$\mathbf{H}_f$ pooled over $[0, 4], (4, 8], (8, 12]$ s
Attention-Weighted Multiscale Aggregation
Deep Latent Token: $\mathbf{z}_{\text{deep}} \in \mathbb{R}^{B \times 64}$
$\mathbf{z}_{\text{deep}}$
Hemodynamic fNIRS
Shape: $(B, 72, 124)$
12.0 s at 10.33 Hz ($t \in [0, 12]$ s)
Pre-Trial Baseline MBLL
fNIRS Conv1D Encoder
Spatial Conv1D ($72 \to 32$)
Temporal Conv1D ($32 \to 64$, $k = 5$)
Tokens: $\mathbf{H}_n \in \mathbb{R}^{B \times 124 \times 64}$ (10.33 Hz)
Q
Entropy-Guided Gated Fusion
Entropy-Guided Feature-Wise Gating Network:
$\mathbf{g}_b = \sigma(\mathcal{F}_g([\mathbf{z}_{\text{deep}}; \mathbf{z}_{\text{shallow}}; H(\hat{\mathbf{p}}_{\text{deep}}); H(\hat{\mathbf{p}}_{\text{shallow}})]))$
$\mathbf{g}_b \in (0, 1)^{64}$, MLP $130 \to 32 \to 64$
Stream DropPath Regularization ($p = 0.15$)
Channel-Wise Convex Blending:
$\mathbf{z}_{\text{fused}} = \mathbf{g}_b \odot \mathbf{z}_{\text{deep}} + (\mathbf{1} - \mathbf{g}_b) \odot \mathbf{z}_{\text{shallow}} \in \mathbb{R}^{B \times 64}$
Linear Readout: $\boldsymbol{\ell} = \mathbf{W}_{\text{out}} \mathbf{z}_{\text{fused}} + \mathbf{b}_{\text{out}} \in \mathbb{R}^{B \times 2}$
Total Trainable Parameters: 63,940 (< 65k)
EEG
fNIRS
Sub-Band & Vascular Slope Stream
Sub-Band EEG Oscillatory Power:
θ (4–8 Hz), α (8–13 Hz), β (13–30 Hz) $\in \mathbb{R}^{B \times 90}$
Delayed Capillary Optical Amplitude & Slope:
Mean Perfusion ($t \geq 3.5$ s) & Gradient $\in \mathbb{R}^{B \times 144}$
Raw Biophysical Vector: $\mathbf{x}_{\text{bio}} \in \mathbb{R}^{B \times 234}$
Biophysical Feature Projection
Low-Rank Bottleneck Dimension Reduction:
Linear($234 \to 32$) → LayerNorm → GELU
Dropout(0.15) → Linear($32 \to 64$)
Shallow Vector: $\mathbf{z}_{\text{shallow}} \in \mathbb{R}^{B \times 64}$
Bottleneck Parameters: 9,568
$\mathbf{z}_{\text{shallow}}$